Department of Computer Applications
(B.C.A)

Syllabus

Regulation for the UG Programmes
Choice Based Credit System (CBCS)
LOCF

(For the Candidate Admitted from the Academic Year 2025 - 2026 onwards)

EDAYATHANGUDY G.S. PILLAY ARTS & SCIENCE COLLEGE, NAGAPATTINAM

(An Autonomous Institution, Affiliated to Bharathidasan University, Tiruchirappalli)

BCA Approved by AICTE, New Delhi – 2(f) and 12(B) Recognized by UGC

Accredited by NAAC with 'A' Grade

B.C.A., Course Structure

(For the Candidates admitted from the Academic year 2025-2026 onwards)

Semester	Part	Course	Title	Instru. Hours / Week	Credit	Exam Hours	Marks		Total
							Int	Extn.	
I	I	Language Course – I (LC) – Tamil/Other Languages	Tamil-I	6	3	3	25	75	100
	II	English Language Course - I (ELC)	English-I	6	3	3	25	75	100
	III	Core Course – I (CC)	Programming in C and Data Structures	6	5	3	25	75	100
		Core Practical - I (CP)	Programming in C Lab	4	4	3	50	50	100
		First Allied Course –I (AC)	Algebra & Calculus	6	4	3	25	75	100
	IV	Value Education	Value Education	2	2	3	25	75	100
	Total			**30**	**21**	-	-	-	**600**
II	I	Language Course – II (LC) - – Tamil/Other Languages	Tamil-II	6	3	3	25	75	100
	II	English Language Course – II (ELC)	English-II	6	3	3	25	75	100
	III	Core Course – II (CC)	Programming in Java	5	5	3	25	75	100
		Core Practical - II (CP)	Programming in Java Lab	4	4	3	50	50	100
		First Allied Course – II (AC)	Numerical Analysis &Statistics	3	2	3	25	75	100
		First Allied Course – III (AC)	Operation Research	4	3	3	25	75	100
		Environmental Studies	Environmental Studies	2	2	3	25	75	100
	Total			**30**	**22**	-	-	-	**700**
III	I	Language Course – III (LC)– Tamil/Other Languages	Tamil-III	6	3	3	25	75	100
	II	English Language Course - III (ELC)	English-III	6	3	3	25	75	100
	III	Core Course – III (CC)	Programming in Python	6	5	3	25	75	100
		Core Practical - III (CP)	Programming in Python Lab	4	4	3	50	50	100
		Second Allied Course – I (AC)	Financial Accounting	6	4	3	25	75	100
		Non Major Elective I	A)Principles of Internet B)Basics of HTML	2	2	3	25	75	100
		Activity Based Course	Health and wellness	-	1	-	20	80	100
	Total			**30**	**22**	-	-	-	**700**

Semester	Part	Course	Title	Instru. Hours/ Week	Credit	Exam Hours	Marks		Total
IV	I	Language Course –IV (LC) - Tamil/Other Languages	Tamil-IV	6	3	3	25	75	100
	II	English Language Course–IV (ELC)	English-IV	6	3	3	25	75	100
	III	Core Course – IV (CC)	Database Management Systems	5	5	3	25	75	100
		Core Practical - IV (CP)	Database Management Systems Lab	4	3	3	50	50	100
		Second Allied Practical – I (AP)	Computer Application in Business with tally lab	3	2	3	50	50	100
		Second Allied Course–II (AC)	Computer Application in Business	4	3	3	25	75	100
	IV	Non Major Elective II	A) Information security & cyber law B) Biometrics	2	2	3	25	75	100
	Total			**30**	**21**	-	-	-	**700**
V	III	Core Course V [CC]	Programming in Php	5	5	3	25	75	100
		Core Course VI [CC]	Computer Network	5	5	3	25	75	100
		Core Course VII [CC]	Operating system	6	5	3	25	75	100
		Core Practical V [CP]	Programming in Php Lab	5	4	3	50	50	100
		Major Based Elective - I	1. Internet of things and its Applications 2. Cloud Computing	5	4	3	25	75	100
		Internship	Internship	-	2	-	20	80	100
	IV	Skill Based Elective I	1.Amazon Web Service 2.Web Designing	2	2	3	25	75	100
		Soft Skills Development	Soft Skills Development	2	2	3	25	75	100
	Total			**30**	**29**	-	-	-	**800**
VI	III	Core Course VIII [CC]	Software Engineering	6	5	3	25	75	100
		Core Course IX [CC]	ASP. NET Programming	6	5	3	25	75	100
		Core Practical VI [CP]	ASP.NET Programming (Lab)	5	4	3	50	50	100
		Major Based Elective - II	1.Artificial Intelligence 2. Network Security	5	4	3	25	75	100
		Project	Project	5	3	-	20	80	100
	IV	Skill Based Elective II	1.Multimedia System 2.Enterprise Resource Planning	2	2	3	25	75	100
	V	Gender Studies	Gender Studies	1	1	3	25	75	100
		Extension Activities	Extension Activities	-	1	-	-	-	-
	Total			**30**	**25**	-	-	-	**700**
Grand Total				**180**	**141**	-	-	-	**4200**

27/06/2025
R.M.LALLI, MCA.,M.Phil.,Ph.D.,
Associate Professor
Department of Computer Science,
Bharathidasan University,

Dr. A. ADHISELVAM
PROFESSOR & HEAD,
DEPT. OF IT
Dr. N.G.P. ARTS AND SCIENCE COLLEGE
Coimbatore-48

27/6/25
Dr.P.PANIMALAR, M.C.A.,M.Phil.,S.E.T.,Ph.D.,
ASSOCIATE PROFESSOR OF COMPUTER SCIENCE
A.V.C. COLLEGE (AUTONOMOUS)
MANNAMPANDAL, MAYILADUTHURAI-609 305.

E. MANIVANNAN,
Chairman - BOS,
E. G. S. Pillay Arts & Science College (A),
Nagapattinam- 611002.

COURSES & CREDITS FOR BCA PROGRAMME

Part	Course	No. of Courses	Mark	INSTRU. HOURS / WEEK	Credits
I	Tamil/Other Language	4	400	24	12
II	English	4	400	24	12
III	Core Theory	9	900	50	45
	Core Practical	6	600	26	23
	First Allied(mathematics)	3	300	13	9
	Second Allied (Financial Accounting & Computer Application in Business)	2	200	10	7
	Second Allied (Tally Practical)	1	100	3	2
IV	NME	2	200	4	4
	SBE	2	200	4	4
	MBE	2	200	10	8
	Value Education	1	100	2	2
	Environmental Studies	1	100	2	2
	Soft Skill Development	1	100	2	2
	Internship	1	100	-	2
V	Project Work	1	100	5	3
	Gender Studies	1	100	1	1
	Extension Activities (Credit only)	1	-	-	1
		43	**4200**	**180**	**140**

a) Basic Tamil I & II for other language students
b) Special Tamil I & II for those who studied Tamil upto 10thor +2 but opt for other languages in degree programme
c) Intership-10 to 15 days

Note:	Internal Marks	External Marks
1. Theory	25	75
2. Practical	50	50

3. Internship	20	80

EDAYATHANGUDY G.S. PILLAY ARTS & SCIENCE COLLEGE, NAGAPATTINAM

(An Autonomous Institution, Affiliated to Bharathidasan University, Tiruchirappalli)

MBA Approved by AICTE, New Delhi – 2(f) and 12(B) Recognized by UGC

Accredited by NAAC with 'A' Grade

B.C.A., Course Structure

(For the Candidates admitted from the Academic year 2025-2026 onwards)

S.No	Semester		Subject Name
1	III	Non Major Elective I	1.Principles of Internet 2.Basics of HTML
2	IV	Non Major Elective II	1.Information security & cyber law 2.Biometrics

S.No	Semester		Subject Name
1	V	Skill Based Elective I	1.Amazon Web Services 2.Web Designing
2	VI	Skill Based Elective II	1.Multimedia System 2.Enterprise Resource Planning

B.C.A., Course Structure
(For the Candidates admitted from the Academic year 2025-2026 onwards)

Sem	Part	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
							Int.	Ext.	Total
I	I	Language Course–I(LC) – Tamil Other Languages	Tamil-I	6	3	3	25	75	100
	II	English Language Course - I(ELC)	English-I	6	3	3	25	75	100
	III	Core Course–I(CC)	Programming in C and Data Structure	6	5	3	25	75	100
		Core Practical-I(CP)	Programming in C lab	4	4	3	50	50	100
		First Allied Course–I(AC)	Algebra & Calculus	6	4	3	25	75	100
	IV	Value Education	Value Education	2	2	3	25	75	100
	Total			**30**	**21**				**600**

Sem	Part	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
							Int.	Ext.	Total
II	I	Language Course–II(LC)-Tamil/Other Languages	Tamil-II	6	3	3	25	75	100
	II	English Language Course–II(ELC)	English-II	6	3	3	25	75	100
	III	Core Course–II(CC)	Programming in Java	5	5	3	25	75	100
		Core Practical-II(CP)	Programming in Java Lab	4	4	3	50	50	100
		First Allied Course–II(AC)	Numerical Analysis &Statistics	4	3	3	25	75	100
		First Allied Course–III(AC)	Operation Research	3	2	3	25	75	100
	IV	Environmental Studies	Environmental Studies	2	2	3	25	75	100
	Total			**30**	**22**				**700**

Sem.	PART	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum marks		
							Int.	Ext.	Total
III	I	Language Course–III(LC) -Tamil/Other Languages	Tamil-III	6	3	3	25	75	100
	II	English Language Course-III (ELC)	English-III	6	3	3	25	75	100
	III	Core Course–III (CC)	Python in Programming	6	5	3	25	75	100
		Core Practical-III(CP)	Python Programming Lab	4	4	3	50	50	100
		Second Allied Course–I(AC)	Financial Accounting	6	4	3	25	75	100
	IV	Non Major Elective I -	A)Principles of Internet B)Basics of HTML	2	2	3	25	75	100
		ACTIVITY BASED COURSE	Health and Wellness	-	**1**	-	-	-	**100**
	Total			**30**	**22**				**700**

Sem.	Part	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
							Int.	Ext.	Total
IV	I	Language Course–IV(LC) - Tamil/Other Languages	Tamil-IV	6	3	3	25	75	100
	II	English Language Course–IV (ELC)	English-IV	6	3	3	25	75	100
	III	Core Course–IV(CC)	Database Management Systems	5	5	3	25	75	100
		Core Practical-IV(CP)	Database Management Systems Lab	4	4	3	50	50	100
		Second Allied Course-II(AC)	Computer Application in Business	4	3	3	25	75	100
		Second Allied Practical– I (AP)	Computer Application in Business with tally lab	3	2	3	50	50	100
	IV	Non Major Elective II-	A)Information security & cyber law B) Biometrics	2	2	3	25	75	100
	Total			**30**	**22**				**700**

Sem	PART	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
							Int.	Ext.	Total
	III	Core Course V[CC]	Programming in PhP	6	5	3	25	75	100
		Core Course VI[CC]	Computer Network	6	5	3	25	75	100
		Core Course VII[CC]	Operating system	5	5	3	25	75	100
		Core Practical V[CP]	Programming in PhP Lab	4	3	3	50	50	100
		Major Based Elective-I	1. Internet of things and its Applications 2. Cloud Computing	5	4	3	25	75	100
		Internship	Internship (10-15 Days During Semester Holidays)	-	2	-	20	80	100
	IV	Skill Based Elective-I	1.Amazon Web Service 2.Web Designing	2	2	3	25	75	100
		Soft Skills Development	Soft Skills Development	2	2	3	25	75	100
		Total		**30**	**28**				**800**

Sem	PART	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
							Int.	Ext.	Total
	III	Core Course VIII[CC]	Software Engineering	6	5	3	25	75	100
		Core Course IX[CC]	ASP. NET Programming	6	5	3	25	75	100
		CorePractical VI[CP]	ASP.NET Programming Lab	5	4	3	50	50	100
		Major Based Elective-II	1.Artificial Intelligence 2.Network Security	5	4	3	25	75	100
		PROJECT	Project	5	3	3	20	80	100
	IV	Skill Based Elective -II	1.Multimedia System 2.Enterprise Resource Planning	2	2	3	25	75	100
	V	Gender Studies	Gender Studies	1	1	3	25	75	100
		Extension Activities	Extension Activities	-	1	-	-	-	-
		Total		**30**	**25**				**700**
		GRAND TOTAL		**180**	**140**				**4200**

20/05/2026

DR.M.LALLI, MCA.,M.Phil.,Ph.D.,
Associate Professor
Department of Computer Science,
Bharathidasan University,
Trichy - 620 023.

20/5/26

Dr. A. ADHISELVAM, Ph.D.
PROFESSOR AND HEAD
Department of Information Technology
Dr. N.G.P. Arts and Science College
Coimbatore - 641 048

T.THANIVANNAN,
Chairman - BOS,
Nagapattinam - 611002.

20/5/26

Dr.P.PANIMALAR, M.C.A.,M.Phil.,S.E.T.,Ph.D.,
ASSOCIATE PROFESSOR OF COMPUTER SCIENCE
A.V.C. COLLEGE (AUTONOMOUS)
MANNAMPANDAL, MAYILADUTHURAI 609 305.

SKILL BASED ELECTIVE–I		
S.NO	**SEMESTER**	**TITLEOFTHEPAPERS**
1.	**V**	1.Amazon Web Services 2.Web Designing

SKILL BASED ELECTIVE–II		
S.NO	**SEMESTER**	**TITLE OF THE PAPERS**
1.	**VI**	1.Multimedia System 2.Enterprise Resource Planning

NON MAJOR ELECTIVE–I		
S.NO	**SEMESTER**	**TITLE OF THE PAPERS**
1.	**III**	1.Principles of Internet 2.Basics of HTML

NON MAJOR ELECTIVE–II		
S.NO	**SEMESTER**	**TITLE OF THE PAPERS**
1.	**IV**	1.Information security & Cyber law 2.Biometrics

LIST OF ALLIED COURSES

First Allied Course I
Mathematics

Second Allied Course II
Financial Accounting

SUMMARY OF CURRICULUM STRUCTURE OF UG PROGRAMMES

S.NO	PART	TYPESOFTHECOURSE	NO.OF COURSE	NO.OF CREDITS	MARKS
1	I	Language Part–I	4	12	400
2	II	English Part–II	4	12	400
3	III	Core Paper	9	45	900
4		Core Practical	6	24	600
5		Allied Paper	5	15	500
6		Allied Practical	1	2	100
7		Major Based Elective Course	2	8	200
8		Project	1	3	100
9		Internship	1	2	100
10	IV	Non-Major Elective	2	4	200
11		Skill Based Elective	2	4	200
12		Environmental Studies	1	2	100
13		Value Education	1	2	100
14		Soft Skill Development	1	2	100
15		Activity based course	1	1	100
16	V	Gender Studies	1	1	100
17		Extension Activities	1	1	100
TOTAL			**43**	**140**	**4200**

*for those who studied Tamil upto 10th +2 (Regular Stream)
+Syllabus for other Languages should be on par with Tamil at degree level #
those who studied Tamilupto10th+2but opt for other languages in degree level under Part I should study special Tamil in Part IV

**Extension Activities shall be outside instruction hours.
Internship are 10 to 15 days.

Note:

	Internal Marks	External Marks
1.Theory	25	75
2.Practical	50	50

3. Separate passing minimum is prescribed for Internal and External marks

FOR THEORY

The passing minimum for CIA shall be 40% out of 25 marks [i.e.10marks] .

The passing minimum for University Examinations shall be 40% out of 75 marks [i.e.30marks]

FOR PRACTICAL

The passing minimum for CIA shall be 40% out of 50 marks [i.e.20 marks]

The passing minimum for University Examinations shall be 40% out of 50 marks [i.e.20marks

PROGRAMME OUTCOMES :

- To recognize the value of life long learning and the role of science in society.
- To identify the contemporary issues and demonstrate the scientific knowledge through various processes to arrive at solutions.
- To collect, assess and make use of the information in both theoretical and experimental science.
- To apply advanced theoretical and/or experimental methods, including the use of laboratory techniques, numerical methods and computer simulations in various fields and practical problems.
- To disseminate the science information effectively both in written or by orally before any audience.

PROGRAMME SPECIFIC OUT COMES :

- To apply the knowledge of Computer Science to identify, analyze problems and to provide effective solution in the area of Computing.
- To develop skills to excel in the fields of Computer Science and its enabled services invarious fields of Government and Private sectors.
- To employ modern computer languages and applications for their successful career, and to create platforms to become an entrepreneur.
- To instill ethical responsibilities, human and professional values and make their contribution to the society.

FIRST YEAR	CORE COURSE–I	COURSECODE: 25BCACC1
SEMESTER I	PROGRAMMING IN C AND DATA STRUCTURE	CREDITS:5

COURSE OBJECTIVES:

- To know about the basics of C Programming.
- To know about Control and Looping Structure .
- To understand Arrays, Pointers and String Processing in C language.
- To know about the basic concepts in Data Structures.

UNITS	CONTENTS	HOURS
I	Basic of C: History of C and its importance – Structure of a C program – Data Types – Constants and Variables – Operators and Expressions – Order of Precedence, Evaluating of Arithmetic Expressions – Type Conversion- Decision Statements: if, if-else, and nested if statements.	20
II	Loops Structures: For Loop, While, Do-while loop – Arrays: - One Dimensional Array, Two-dimensional Arrays, Character Arrays and Strings – Functions: Function with arrays-Function with decision and looping statements- Recursion.	20
III	Pointers: Introduction – Pointer Expressions – Chain of Pointers –Pointers and Arrays–ArrayofPointers–Pointersasfunctionarguments– Functionsreturning Pointers – Pointers to Functions – Function pointer – Structures - declaration, initialization, Array of Structures –Pointer to structures, Structures and functions – Type def, Enumerated data types, Unions.	20
IV	Strings Processing, Standard string library functions–Files: introduction and files functions–Writing and reading in Text mode–Simple application: Display the contents of a file. Write data to a file. Append data to an existing file – Simple application: Display the contents of a file. Write data to a file. Append data to existing file–File IO–Reading and writing structures.	15
V	Stack: LIFO concept, Stack operations, Array implementation of stack– Queue: FIFO concept, Queue operations, Array implementation of queue– Singly Linked List: concepts operations–Doubly Linked List: concepts, operations–Trees: General trees, Binary trees.	15

TEXTBOOKS:

1. E.Balagurusamy, "Programming in ANSI C", Tata McGraw Hill, NewDelhi, Seventh Edition, 2016.
2. E. Horowitz, S. Sahni and Susan Anderson Freed, "Fundamental Data Structures in C", 2ed, Orient Black Swan Publisher, 2009.
3. Byron S. Gottfried, "Programming with C", Schaum's Outline Series, Tata-McGraw Hill Edition, New Delhi, 1991.
4. E.Karthikeyan,"ATextbook on C Fundamentals, DataStructures and Problem Solving", Prentice-Hall of India Private Limited, New Delhi, 2008.
5. Yashavant Kanetkar, "Let us C", BPB Publications, Tenth Edition, NewDelhi, 2010.
6. Szuhay, Jeff, and Szuhay, Jeff, "Learn C Programming: A Beginner's Guide to Learning C Programming the Easy and Disciplined Way", Packt Publishing, 2020.
7. Jena, SisirKumar, and Jena, SisirKumar,"C Programming :Learn to Code", CRC Press, 2021.

REFERENCE BOOKS:

1. E.Karthikeyan,"ATextbook on C Fundamentals, DataStructures and Problem Solving", Prentice-Hall of India Private Limited, New Delhi, 2008.
2. Yashavant Kanetkar, "Let us C", BPB Publications, Tenth Edition, NewDelhi, 2010.

WEBRESOURCES:

1. https://www.tutorialspoint.com/cprogramming/index.htm
2. https://www.w3schools.in/data-structures/intro

COURSEOUTCOMES:

Upon successful completion of this course the students would be able to:

CO #	CO Statement	K-Level
C01	Summarize the basic knowledge to develop C programs	K2
C02	Manipulate Looping,arrays and functions	K3
C03	Apply and write programs for solving real world problems	K4
C04	Create open, read, manipulate, write and close files.	K5
C05	Understand the basic concepts in data structures.	K2

FIRST YEAR	CORE PRACTICAL-I	COURSECODE: 25BCACP1
SEMESTER I	PROGRAMMING IN C LAB	CREDITS:3

COURSE OBJECTIVES :

- To provide the necessary knowledge and practical training for step by step computer program development and to present the basic concepts in C programming language and to prepare the students to write modular and readable C Programs.

- Also the Lab Course implements the essential concepts like data types.

- To analyze and how to use such knowledge for later processing with the help of files and aims to train the students to write working programs to solve problems

LISTOFPROGRAMS:

1. Solution of a Quadratic Equation(all cases).
2. Sum of Series(sine, cosine).
3. Ascending and descending order of numbers using Arrays(Use it to find Largest and Smallest Numbers).
4. Sorting of names in Alphabetical order.
5. Matrix operations (Addition, Subtraction, Multiplication–using functions.)
6. Finding factorials, generating Fibonacci Numbers using recursive functions.
7. String manipulations using string functions (string length, string comparison, string copy, palindrome checking, counting words and lines in strings).

   Develop a program to create an employee details using structure and prepare a pay slip.
8. Creation and processing of Sequential files for Mark list preparation.
9. Write a program to implement the List using arrays

COURSE OUTCOMES:

CO #	CO Statement	K-Level
C01	Understanding how problems are posed and how they can be analyzed for obtaining solutions.	K1
C02	Understanding the fundamentals of C programming.	K2
C03	Learning of sequencing, branching, looping and decision making statements to solve scientific and engineering problems.	K3
C04	Implementing different operations on arrays and creating and using of functions to solve problems.	K4
C05	Ability to design and implement different types of file structures using standard methodology.	K5

ALLIED MATHEMATICS

for

B.C.A, B.Sc., Artificial Intelligence and Machine Learning, B.Sc., Computer Science, B.Sc., Information Technology Programmes

First Year	ALLIED COURSE - I			Credit: 4
Semester - I	ALGEBRA AND CALCULUS			Subject Code:
Instructional Hours per week	Lecture	Tutorial	Practical	Total
	6	-	-	6

COURSE OBJECTIVES:

- To lean the basic concepts in the integration.
- To train the students to solve the problems in Theory of Equations.
- To provide knowledge about the matrix, differentiation.

Units	CONTENT	HOURS
Unit I	**Theory of Equations:** Relation between roots & coefficients – Transformations of Equations – Diminishing, Increasing & multiplying the roots by a constant- Forming equations with the given roots.	18
Unit II	**Matrices:** Singular matrices – Inverse of a non-singular matrix using adjoint method - Rank of a Matrix – Consistency - Characteristic equation, Eigen values, Eigen vectors.	18
Unit III	**Differentiation:** Maxima & Minima – Concavity, Convexity – Points of inflexion - Partial differentiation – Euler's Theorem - Total differential coefficients (proof not needed) –Simple problems only.	18
Unit IV	**Integration :** Evaluation of integrals of types 1) $\int \frac{px+q}{ax^2+bx+c}dx$ 2) $\int \frac{px+q}{\sqrt{ax^2+bx+c}}dx$ - Evaluation using Integration by parts.	18
Unit V	**Differential Equations:** Second order of types $(aD^2 + bD + c)y = F(x)$ where a, b, c are constants and $F(x)$ is one of the following types (i) e^{kx} (ii) $si(kx)$ *or* $cos(kx)$ (iii) x^n, n being an integer (iv) $e^{kx} f(x)$.	18

COURSE OUTCOMES		COGNITIVE LEVEL
CO1	Train the students to solve the problems in theory of equations.	K1
CO2	Know about the origin of matrices and find the solutions of Eigen problems.	K2
CO3	Get exposed the basic concepts of differentiation and integration	K3
CO4	Acquire the knowledge about integral equations.	K4
CO5	Acquire the knowledge about differential equations.	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create.		

TEXTBOOKS	
1.	T.K. Manickavasagam Pillai & others, Algebra, Volume I, S.V Publications, 1985 Revised Edition (Units I, II).
2.	S. Narayanan, T.K. Manicavachagam Pillai, Calculus, (Volume II), S. Viswanathan Pvt Limited, 2003. (Units III, IV and V).
REFERENCE BOOKS	
1.	M.L. Khanna, Differential Calculus, Jaiprakashnath and Co., Meerut-2004
WEB RESOURCES	
1.	http://mathforum.org
2.	http://ocw.mit.edu/ocwweb/Mathematics

FIRST YEAR	CORE COURSE–II	COURSE CODE: 25BCACC2
SEMESTER II	PROGRAMMING IN JAVA	CREDITS: 5

COURSE OBJECTIVES:

- To acquire the programming skills with java.
- To implement the object oriented concepts with java language
- To learn the art of GUI programming with Applet.

UNITS	CONTENTS	HOURS
I	Foundation, Essentials, Control Statement and Classes &Objects. Stage of Java–origin of Java–challenges-features-Object-Oriented Programming; Java Essentials: Elements - API - variables - primitive data types –String Class - operators –combined assignment operators - conversion –scope – comments - keyboard input; Control Statements: if, if-else, nested if& if-else-if statements – logical operators – comparison – conditional operator – switch– increment and decrement – while, do-while &for loops – nested loops – break and continue; Classes and Objects: classes and objects - modifiers - passing arguments– constructors - package& import - static class members –method overloading–constructoroverloading–returningobjects–thisvariable–recursion– nested & inner classes–abstract classes & methods.	15
II	Arrays, String Handling, Inheritance, Interface and Packages. Introduction–processing array–two Dimensional Arrays- Arrays with Three or More Dimensions- String Handling : Stringclass–concatenation–comparison–substring–methods–StringBuffer, String Builder & StringTokenizer classes; Inheritance: basics –inheriting and overriding superclassmethods–callingsuperclassconstructor–polymorphism–inheritfromdifferentclasses –abstractclasses–Interfaces:Basics—multilevelinterface–Packages.	15
III	Exception Handling, I/O and File Handling and Multithreading. Introduction-throw Statement– throw Clause - Multithreading: Introduction – threads — life cycle – joining a thread – scheduler & priority – synchronization – thread control –File and I\O Streams: file Class –streams –byte streams–Random Access File	15
IV	Applet and GUI Part I .Fundamentals– applet class–lifecycle–steps for applet program – graphics – event handling; GUI I:GUI – creating windows – dialog boxes – layout managers – AWT component classes – applications of AWT controls.	15
V	GUI Part II and Java Database Connectivity Event handling Swing Concepts: Swing controls–application using Swing and AWT; Java Database Connectivity: types of drivers–JDBC architecture – JDBC classes & interfaces–steps in JDBC applications–creating a new Data base and table with JDBC.	15

TEXTBOOKS:

1. S. Sagayaraj, R. Denis, P. Karthik& D. Gajalakshmi, "Constructive Java Programming",Universities Press,2021.
2. E.Balagurusamy,"ProgrammingwithJAVA",TataMcGrawHill,NewDelhi,2019.
3. C.Muthu, "Programming with JAVA", Vijay Nicole Imprints Private Limited, Chennai, Second Edition, 2011.

REFERENCE BOOKS:

1. BruceEckel,ChuckAllison,"Thinking in Java", Prentice Hall Publications, 2006
2. Malina Pronto,"Java: How To Learn Java Programming: How To Improve Your Java Coding In 2020/2021:5 Programming Languages To Learn For Beginners In Tech", Independently Published,2020.
3. Nick Samoylov, "Learn Java 12 Programming:A Step-by-step Guide to Learning Essential Concepts in Java", Packt Publishing, 2019.

WEB RESOURCES:

https://www.javatpoint.com/java-tutorial

COURSE OUTCOMES:

Upon successful completion of this course the students would be able to:

CO #	CO Statement	K-Level
C01	Remember the fundamentals of programming such as variables, Conditional statements and iterative execution statements.	K1
C02	Understanding the OOPs concepts and String	K2
C03	Apply the concepts of Exception Handling.	K3
C04	Implement the concepts of Multithreading and File Streams.	K4
CO5	Analyze the concepts of applet programming, graphics programming, Java swing..	K5

FIRST YEAR	CORE PRACTICAL- II	COURSE CODE: 25BCACP2
SEMESTER II	PROGRAMMING IN JAVA LAB	CREDITS:3

COURSE OBJECTIVES:

- To understand the basics of JAVA programs and their execution.
- To learn concepts like inheritance, packages and interfaces.
- To understand the lifecycle of the applets, database connectivity and their functionality.

LIST OF PROGRAMS:

1. Write a program to sort the given numbers using arrays.
2. Write a program to implement the FIND and REPLACE operations in the given text.
3. Write a program to implement a calculator to perform basic arithmetic Operations, doing with constructers
4. Write a program to find the student's percentage and grade using command line arguments.
5. Write a program to draw circle or triangle or square using polymorphism and inheritance.
6. Implement multiple inheritance concepts in java using interface , you can choose your own example of a company or education institution or a general concept which requires the use of interface to solve a particular problem.
7. Write a program to create threads and perform operations like start, stop, suspend, resume
8. Write a program to develop an applet to play multiple audio clips using multithreading.
9. Write a program to retrieve employee data from a file
10. Write a program to retrieve student data from a Database

COURSE OUTCOMES:

CO #	CO Statement	K-Level
C01	Remember the fundamentals of Java programming language	K1
C02	Understand the basics of Java programming, multi-threaded programs andException handling	K2
C03	Analyze and use Java in a variety of applications.	K3
C04	Write and debug a software application developed using the Java Programming language.	K4
C05	Ability to design application programs using Java	K5

First Year	ALLIED COURSE - II			Credit: 3
Semester - II	NUMERICAL ANALYSIS AND PROBABILITY			Subject Code:
Instructional Hours per week	Lecture	Tutorial	Practical	Total
	4	-	-	4

COURSE OBJECTIVES:

- To train the students in the numerical problems.
- To train the students in solving statistical problems.
- To learn knowledge about an algebraic and transcendental equations.

Units	CONTENT	HOURS
Unit I	**Algebraic & Transcendental equations:** Bisection Method, Newton Raphson Method, Iteration method - Finite differences –Forward, Backward differences.	12
Unit II	Numerical differentiation - Numerical Integration using Trapezoidal rule and Simpson's first & second rules (proof not needed) - Solutions to Linear Systems-Gauss elimination method.	12
Unit III	**Numerical solution of ODE:** Solution by Taylor Series Method, Euler's Method, Runge - Kutta 2nd order method.	12
Unit IV	Arithmetic Mean – Geometric Mean – Harmonic Mean - Median, Mode, Standard Deviation - Quartile Deviation.	12
Unit V	Correlation and Regression – Properties of Simple Correlation and regression coefficients – Simple Numerical Problems only.	12

COURSE OUTCOMES		COGNITIVE LEVEL
CO1	Solve algebraic and transcendental equations.	K1,K2
CO2	Understand the list of basic ideas of Numerical Methods and Statistics.	K2,K3
CO3	Solve differential and integral equations numerically.	K3
CO4	Analyze the accuracy and graphical representation of statistical data	K4
CO5	Understand the correlation and regression coefficients	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create.		

TEXTBOOKS	
1.	S.S.Sastry, Numerical Analysis (Unit 1 , 2 , 3)
2.	Gupta.S.C & Kapoor, V.K, Fundamentals of Mathematical Statistics, Sultan Chand & sons, New Delhi -1994 (Units 4 & 5)
REFERENCE BOOKS	
1.	M.K. Jain, S.R.K. Iyengar and R.K. Jain, Numerical Methods for Scientific and Engineering Computation, New Age International Private Limited, 1999.
2.	C.E. Froberg, Introduction to Numerical Analysis, II Edn., Addison Wesley, 1979.
WEB RESOURCES	
1.	http://mathforum.org
2.	https://nptel.ac.in/courses/111/105/111105090/

First Year	ALLIED COURSE - III			Credit: 2
Semester - II	OPERATIONS RESEARCH			Subject Code:
Instructional Hours per week	Lecture	Tutorial	Practical	Total
	4	-	-	4

COURSE OBJECTIVES:

- To train the students to solve assignment problems, transportation problems.
- To train the students in network problems.
- To make students solve real life problems in Business and Management.

Units	CONTENT	HOURS
Unit I	**Operations Research:** Introduction - Basics of OR – OR & decision making – Role of Computers in OR - Linear programming formulations & graphical solution of two variables.	12
Unit II	Canonical & standard forms of LPP - Simplex Method : Simplex Method for < , = , > constraints – Charne's method of penalties.	12
Unit III	**Transportation problem:** Transportation algorithm –Degeneracy algorithm – Degeneracy in Transportation Problem , Unbalanced transportation problem- Assignment algorithm.	12
Unit IV	**Sequencing Problem:** Processing of n jobs through two machines – Processing of n jobs through 3 machines – processing of two jobs through m machines.	12
Unit V	**Networks:** Network – Fulkerson's rule - measure of activity – PERT computation – CPM computation.	12

COURSE OUTCOMES		COGNITIVE LEVEL
CO1	Recognize and relate LPP and solving LPP using graphical method.	K1
CO2	Compute Simplex Algorithm and Big-M Method of LPP.	K2
CO3	Explain Transportation problem and Evaluate its initial basic feasible solution and solve assignment problem using Hungarian algorithm.	K3
CO4	Sequencing the jobs to be carried out based on Cost Optimization.	K4
CO5	Describe and Construct Network and compute PERT and CPM.	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create.		

TEXTBOOKS	
1.	Manmohan & Gupta , Operations Research, Sultan Chand Publishers, New Delhi.
REFERENCE BOOKS	
1.	Prem Kumar Gupta and D.S. Hira, Operations Research : An Introduction, S. Chand and Co., Ltd. New Delhi.
2.	Hamdy A. Taha, Operations Research (7th Edn.), McMillan Publishing Company, New Delhi, 1982.
WEB RESOURCES	
1.	https://www.britannica.com/topic/operations-research
2.	http://ebooks.lpude.in.operationsresearch

SECOND YEAR	CORE COURSE–III	COURSE CODE: 25BCACC3
SEMESTER III	PYTHON IN PROGRAMMING	CREDITS: 5

COURSE OBJECTIVES:

- To make students understand the concepts of Python programming
- To apply the OOPs concept in Python programming
- To make the students learn best practices in Python programming

UNITS	CONTENTS	HOURS
I	Basics of Python Programming: Features of Python -History of Python- Literal Constants-Variables and Identifiers–Data Types- Input Operation- Comments– Reserved Words- Indentation- Operators and Expressions –Other Data Types- Type Conversion	15
II	Decision Control Statements: Selection/Conditional Branching statements: if, if-else, nested if and if-elifelse statements. Basic Loop Structures / Iterative Statements: while loop, for loop- Nested Loops- The break Statement- The continue Statement.	15
III	Functions and Modules: Function Definition – Function Call: Function Parameters – Variable Scope and Lifetime: Local and Global VariablesUsing the Global Statement- Resolution of Names. The return Statement. More on Defining Functions: Required Arguments, Keyword Arguments, Default Arguments and Variable Length Arguments. Python Strings: Strings are Immutable- Built-in String Methods and Functions – Comparing Strings. Modules: The from…import statement- Name of Module – The dir() function – Modules and Namespace.	15
IV	Lists: Access values in List-Updating values in Lists Nested lists -Basic list operations-List Methods. Tuple: Creating, Accessing, Updating and Deleting Elements in a tuple – Nested tuples. Dictionaries: Creating a dictionary, Accessing values, Modifying an Entry -Deleting items – Built-in Dictionary Functions and Methods - Difference between a List and a Dictionary.	15
V	File Handling: Types of files in Python - Opening and Closing files- Reading and Writing files: write() and writelines() methods- append() method – read() and readlines() methods – Splitting words –File Positions.	15

TEXTBOOKS:

- Reema Thareja. (2017), Python Programming using problem solving approach, 1 st Edition, Oxford University Press.

REFERENCE BOOKS:

- Dr. R. Nageswara Rao. (2017), Core Python Programming, 1 st Edition, Dreamtech Publishers.
  2. VamsiKurama.(2017), Python Programming: A Modern Approach,1st Edition, Pearson Education.
- Mark Lutz. (2013), Learning Python, Fifth Edition, Orielly.
- Adam Stewarts. (2017), Python Programming, Online.
- Fabio Nelli. (2015), Python Data Analytics,1st Edition, APress.
- Kenneth A. Lambert. (2019), Fundamentals of Python – First Programs, 2nd Edition, CENGAGE Publication.

WEB RESOURCES:

1. https://www.programiz.com/python-programming
2. https://www.guru99.com/python-tutorials.html
3. https://www.w3schools.com/python/python_intro.asp
4. https://www.geeksforgeeks.org/python-programming-language/
5. https://en.wikipedia.org/wiki/Python_(programming_language)

COURSE OUTCOMES:

Upon successful completion of this course the students would be able to:

CO #	CO Statement	K-Level
CO1	Recall the fundamental concepts of Python	K1
CO2	Demonstrate the problem-solving approach using Python statements	K2
CO3	Construct the Python program using functions and modules	K3
CO4	Analyze the Python programming concepts to develop programs	K4
CO5	Develop a Python program to solve real-time problems	K5

FIRST YEAR	CORE PRACTICAL- III	COURSE CODE: 25BCACP3
SEMESTER III	PROGRAMMING IN PYTHON LAB	CREDITS:3

COURSE OBJECTIVES:

- To provide programming knowledge in Python.
- To create loops and decision statements in Python.
- To build and package Python modules for reusability.
- To read and write files in Python.

LIST OF PROGRAMS:

1. To Implement Python Program using variables, constants, I/O statements .
2. To Implement Python Program using Operators .
3. To Implement Python Program using Conditional Statements.
4. To Implement Python Program using Loops.
5. To Implement Python Program using Functions.
6. To Implement Python Program using Recursion.
7. To Implement Python Program using Arrays.
8. To Implement Python Program using Strings.
9. To Implement Python Program using Tuples.
10. To Implement Python Program for File Handling.

COURSE OUTCOMES:

CO #	CO Statement	K-Level
C01	Recall the fundamental concepts of Python	K1
C02	Identify the problem and solve using Python programming techniques.	K2
C03	Identify suitable programming constructs for problem solving.	K3
C04	Analyze various concepts of Python language to solve the problem in an efficient way.	K4
C05	Develop a Python program for a given problem and test for its correctness.	K5